In JEE Advanced 2016, FIITJEE Students bag 36 in Top 100 AIR, 75 in Top 200 AIR, 183 in Top 500 AIR. 3541 Students from Long Term Classroom/ Integrated School Program & 4423 Students from All Programs have qualified in JEE Advanced, 2016

FIITJEE

JEE (Advanced), 2017

ALL INDIA TEST SERIES

FULL TEST – II

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 234**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 – 12) contains 4 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Section-A (13 to 18) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 – 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

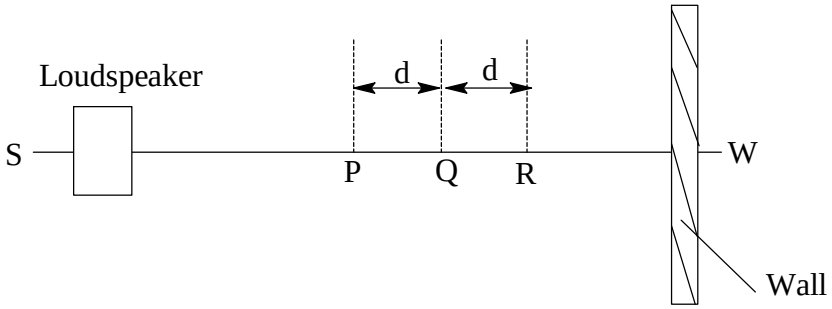
This section contains **08 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. A glass tube of 1.0 meter length is filled with water. The water can be drained out slowly at the bottom of the tube. If a vibrating tuning fork of frequency 500Hz is brought at the upper end of the tube and the velocity of sound is 330m/s then the total number of resonances obtained will be

(A) 4 (B) 3

(C) 2 (D) 1

2. A loudspeaker emits sound of frequency f. The sound waves are reflected from a wall. The arrangement is shown below.



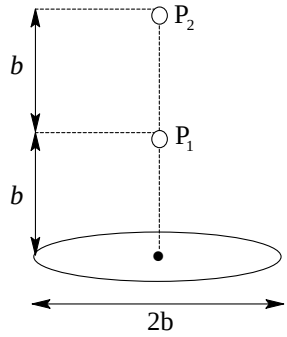
When a microphone is moved along the line SW, minimum loudness of sound is detected at points P, Q and R. There are no other minima between these points. The separation of the minima is d. The speed of the sound wave is :

(A) $\frac{1}{2}fd$ (B) $4fd$

(C) fd (D) $2fd$

Space for rough work

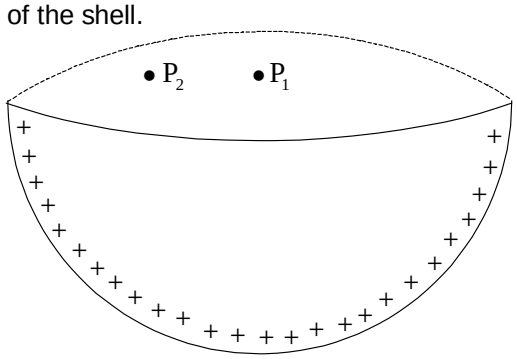
3. Consider a uniformly charged wire that has the form of a circular loop with radius b. Consider two points on the axis of the loop. P_1 is at a distance b from the loop's center, and P_2 is at a distance 2b from the loop's center. The potential V is zero, very far away from the loop. At P_1 and P_2 the potentials are V_1 and V_2, respectively. What is V_2 in terms of V_1?

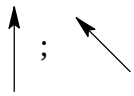
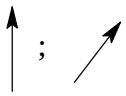


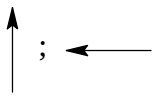
(A) $2V_1$ (B) $\frac{V_1}{2}$

(C) $\sqrt{\frac{5}{2}}V_1$ (D) $\sqrt{\frac{2}{5}}V_1$

4. Consider a uniformly charged hemispherical shell shown below. Indicate the directions (not magnitude) of the electric field at the central point P_1 and an off-centre point P_2 on the drumhead of the shell.

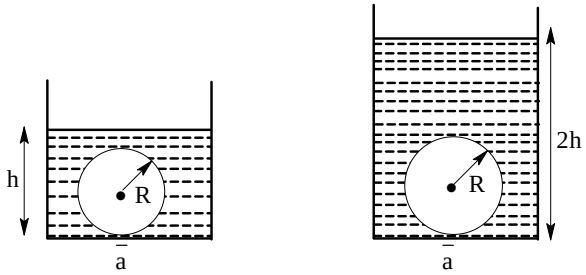


(A)  (B) 

(C)  (D) 

Space for rough work

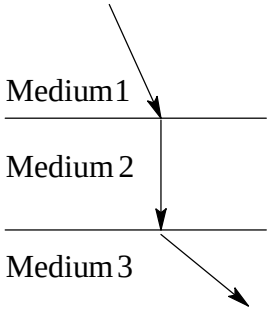
5. Two identical cylinders have a hole of radius a (a<<R) at its bottom. A ball of radius R is kept on the hole and water is filled in the cylinder such that there is no water leakage from bottom. In case-1 water is filled upto height h and in second case it is filled upto height 2h. If F_1 is force by liquid on sphere in case-1 and F_2 is force by liquid on sphere in case-2 then.



(A) $F_1=F_2=0$ (B) $F_1>F_2$

(C) $F_2>F_1$ (D) $F_1=F_2\neq 0$

6. The diagram below shows the path taken by a monochromatic light ray travelling through three media. The symbols v_1, λ_1 and f_1 represent the speed, wavelength, and frequency of the light in Medium 1, respectively. Which of the following relationships for the light in the three media is true.

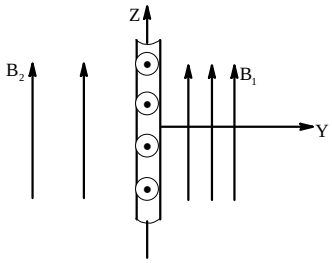


(A) $\lambda_1<\lambda_2<\lambda_3$ (B) $v_1<v_3<v_2$

(C) $f_1<f_3<f_2$ (D) $\lambda_2<\lambda_1<\lambda_3$

Space for rough work

7. Figure shows a thin metal sheet in the plane y = 0, for which the current of constant density flows in the positive x-direction. It is in a constant homogeneous magnetic field of value = (0, 0, B_0). As a result of superposition of magnetic fields in region y > 0, the induction field B_1 = (0, 0, B) and in y < 0 is B_2 = (0, 0, B_2) where $B_1 > B_2$. Specify the correct statement:



(A) $B_0 = \frac{(B_1 - B_2)}{2}$ (B) $B_0 = \frac{(B_1 + B_2)}{2}$

(C) $B_0 = B_1 + B_2$ (D) $B_0 = B_1 - B_2$

8. At ordinary temperature the molecules of an ideal gas have only translational and rotational energy. At high temperature, they may also have vibrational energy. As a result of this at high temperature.

(A) $C_V = \frac{3R}{2}$ for monoatomic gas (B) $C_V > \frac{3R}{2}$ for monoatomic gas

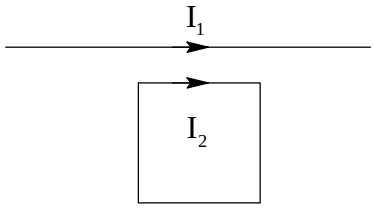
(C) $C_V < \frac{5R}{2}$ for Diatomic gas (D) $C_V > \frac{5R}{2}$ for Diatomic gas

Space for rough work

Multiple Correct Choice Type

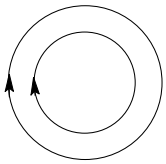
This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

9. A long straight wire carries a steady current I_1. Nearby is a rectangular loop that carries a steady current I_2. The directions of the two currents are shown in the figure.



Which statement is/are false?
(A) The loop is attracted to the wire
(B) There is no net force on the loop from the wire
(C) The loop is attracted to the wire if $I_1 > I_2$; otherwise it is repelled
(D) The loop is repelled from the wire if $I_1 > I_2$; otherwise it is attracted

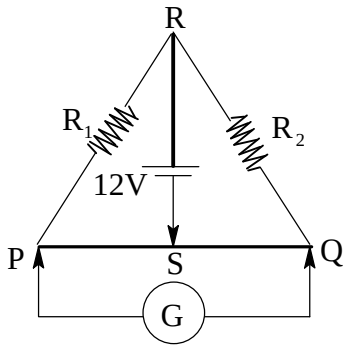
10. Two concentric, coplanar, circular loop of wire, with different diameter carry current in the same sense as shown in the figure. Which of the following statement(s) is (are) correct?



(A) The magnetic force exerted by the outer loop on a short portion of the inner loop is radially outward
(B) The magnetic force exerted by the outer loop on a short portion of the inner loop is radially inward
(C) The net magnetic force exerted by the outer loop on a whole inner loop is non zero and is radially outward
(D) The net magnetic force exerted by the outer loop on a whole inner loop is zero

Space for rough work

11. In the circuit shown $R_1 - R_2 = 10\Omega$ and resistance per unit length of wire $PQ = 1\Omega / cm$ and length $PQ = 10cm$. If R_2 is made 20Ω the to get zero deflection in galvanometer. S is midpoint of wire PQ?



(A) The jockey at P can be moved towards right 2 cm.
(B) The jockey at Q can be moved towards right 2 cm.
(C) The jockey at S can be moved towards left a distance 5/3 cm.
(D) The jockey at all positions fixed and R_1 should be made 20

12. A constant external torque acts for a very brief period t on a rotating system having moment of inertia I.
(A) The angular momentum of the system will change by $\tau \Delta t$
(B) The angular velocity of the system will change by $\frac{\tau \Delta t}{I}$
(C) If the system was initially at rest, it will acquire rotational kinetic energy $\frac{(\tau \Delta t)^2}{2I}$
(D) The kinetic energy of the system will change by $\frac{(\tau \Delta t)^2}{2I}$

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 13 & 14

If the velocity of a charged particle moving through a uniform magnetic field has a component parallel to that field the particle will move in a helical path as shown in figure 1. The net force F on the particle can be described by the equation.

$$F = qvB\sin\theta$$

Where q is the charge on the particle, v is the velocity of the charge, B is the magnetic field strength, and θ is the angle between the velocity and the magnetic field. The pitch p of the helix is the distance between adjacent turns. The radius r of the helix is determined by the component of velocity of the charge perpendicular to the magnetic field and is independent of the component of the velocity parallel to the magnetic field.

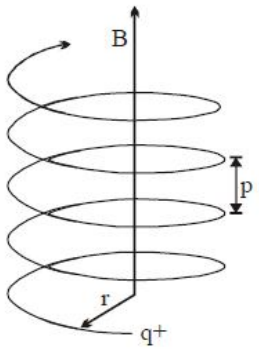


Figure 1

A nonuniform magnetic field that is stronger at its ends than at its middle can trap a charged particle by reflecting it back and forth between ends. Such a situation is shown in figure 2. In a similar phenomenon electrons and protons above the atmosphere are reflected back and forth between the north and south poles of the earth's magnetic field forming the two Van Allen belts high above the atmosphere. Occasionally a solar flare shoots additional electrons and protons into the Van Allen belts. This pushes the electrons to a lesser height. The repulsion due to additional electrons drives the charged particles along the earth's magnetic field lines into the atmosphere where they collide with the air molecules forcing them to a higher energy level emitting photons. Oxygen atoms emit green light and nitrogen atoms emit pink. This light forms the curtain of lights in the sky known in the northern hemisphere as the aurora borealis

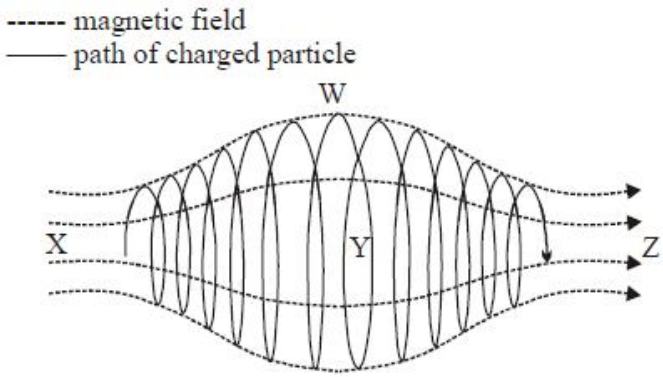


Figure 2 : Path of a charged particle trapped in a changing magnetic field

13. If the magnetic field in figure-2 represents a portion of the Earth's magnetic field, at which labelled point would you except to find the aurora borealis?
(A) W (B) X
(C) Y (D) Z

14. The electrons and protons caught in the Van Allen belt reflect from one pole to the other. Looking up from the North pole, electrons rotate counterclockwise. Protons rotate.
(A) Counterclockwise both when looking up from the North Pole or when looking up from the South Pole.
(B) Clockwise both when looking up from the North Pole of when looking up from the South Pole.
(C) Clockwise when looking up from the North Pole but counterclockwise when looking up from the South Pole.
(D) Counterclockwise when looking up from the North Pole but clockwise when looking up from the South Pole.

Paragraph for Questions 15 & 16

A piano creates sound by gently striking a taut wire with a soft hammer when a key on the piano is pressed. All piano wires in a given piano are approximately the same length. However, each wire is tied down at two points, the bridge and the agraffe. The length of the wire between the bridge and the agraffe is called the speaking length. The speaking length is the part of the wire that resonates. The point of the wire struck by the hammer is displaced perpendicularly to the wire's length. A standing wave is generated by the hammer strike, where □ is the velocity T is the tension in the wire, and μ is the mass per unit length of the wire.

$$v = \sqrt{\frac{T}{\mu}}$$

Tuning a piano involves adjustment of the tension in the wires until just the right pitch is achieved. Correct pitch is achieved by listening to the beat frequency between the piano and a precalibrated tuning fork.

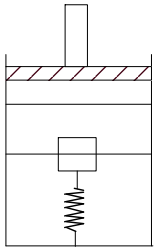
15. A piano note is compared to a tuning fork vibrating at 440 Hz. Three beats per second are discerned by the piano tuner. When the tension in the string is increased slightly, the beat frequency increases. What was the initial frequency of the piano wire?
(A) $434\,Hz$ (B) $437\,Hz$
(C) $443\,Hz$ (D) $446\,Hz$

16. Sound waves move through air at approximately 340 m/s. A piano wire with a 90 cm speaking length resonates at a frequency of 360 Hz. What is the wavelength of the resulting sound wave ?
(A) $0.94\,m$ (B) $1.06\,m$
(C) $4\,m$ (D) $1.8\,m$

Space for rough work

Paragraph for Questions 17 & 18

The figure shows a cylindrical container, fitted with a piston. It contains two immiscible incompressible liquids with densities ρ_1 and ρ_2. A block floats in equilibrium in the position shown.



17. Suppose the spring is initially in compressed state. Now, if the piston is moved downward slowly then
(A) Spring will become more compressed.
(B) Spring will become less compressed.
(C) The compression in spring will remain same as before.
(D) The compression will depends on ρ_1 and ρ_2.

18. Suppose the spring is initially in compressed state. Now, if the whole system is moved with some acceleration upward then
(A) Spring will become more compressed.
(B) Spring will become less compressed.
(C) The compression in spring will remain same as before.
(D) The compression will depend on ρ_1 and ρ_2 .

Space for rough work

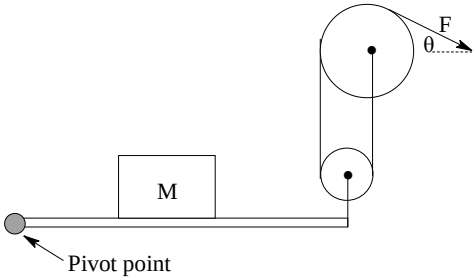
SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. A uniform rod of length l = 1m is free to move and rotate in gravity-free space. When an impulse is given to one end of the rod, perpendicular to its length, its centre of mass moves with velocity v = 1 m/s. What will be its angular velocity (in rad/s) about its centre of mass?

2. A pulley system is attached to a massless board as shown below. The board pivots only at the pivot point. A 10 kg mass M sits exactly in the middle of the board.



If the angle θ is 60°. The force F (in N) necessary to hold the board in the position shown is $5\times x$, find x?

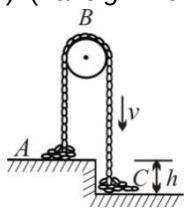
3. The peak emission from a black body at a certain temperature occurs at a wavelength of 6000 Å. On increasing its temperature, the total radiation emitted is increased 16 times. These radiations are allowed to fall on a metal surface. Photoelectrons emitted by the peak radiation at higher temperature can be bought to rest by applying a potential equivalent to the excitation potential corresponding to the transition for the level n = 4 to n = 2 in the Bohr's hydrogen atom. The work function of the metal is given by $\frac{\alpha}{100}eV$ where α is the numerical constant. The value of α is found $53\times x$, find x?

[Take: he = 12420 eV-Å]

Space for rough work

4. A positively charged particle starts at rest 25 cm from a second positively charged particle which is held stationary throughout the experiment. The first particle is released and accelerates directly always from the second particle. When the first particle has moved 25 cm, it has reached a velocity of $10\sqrt{2}$ m/s. The maximum velocity (in m/s) that the first particle will reach is $10\times x$, find x?

5. The ends of a chain lie in piles at *A* and *C*. When given an initial speed 10 m/s, the chain keeps moving freely at that speed over the fixed pulley *B*. Neglecting friction, the required value of *h is* $x\times 5$, then find the value of x (in m). (Take *g* = 10 m/s^2):



Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **08 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. The vapour pressure lowering of a 0.1 M aqueous solution of non – electrolyte at $75°C$ is $\left(\Delta H_v^0 \text{ of water}=9720\,\text{cal/mol and antilog}\,0.407=2.55\right)$

(A) 0.54 torr (B) 0.78 torr
(C) 0.02 torr (D) 0.22 torr

2. The C – C single bond distance is $1.54\,\text{Å}$. The minimum distance between the terminal cartons in propane is $\left(\sin 54°45' = 0.8136\right)$

(A) Greater than $3.08\,\text{Å}$ (B) Equal to $3.08\,\text{Å}$
(C) $2.51\,\text{Å}$ (D) Can't be calculated

3. A sample of aqueous solution of acetic acid $\left(0.5\,\text{mol}\,CH_3COOH\,\text{in}\,10\,\text{mol of}\,H_2O\right)$ is mixed with a sample of aqueous solution of $NaOH\left(0.5\,\text{mol NaOH in}\,10\,\text{mol of}\,H_2O\right)$. The freezing point of the solution is ($K_{f,\,H2O}$=1.86)

(A) $-5°C$ (B) $-5.16°C$
(C) $0°C$ (D) $5.16°C$

4. Hydrogen peroxide can be prepared by the successive reactions

$$2NH_4HSO_4 \rightarrow \left(NH_4\right)_2 S_2O_8$$

$$\left(NH_4\right)_2 S_2O_8 + 2H_2O \rightarrow 2NH_4HSO_4 + H_2O_2$$

The first is an electrolytic reaction, and the second a steam distillation.
What current would have to be used in the first reaction to produce enough intermediate to yied 100 g pure H_2O_2 per hour? Assume 50% current efficiency.

(A) 158 A (B) 315 A
(C) 102 A (D) 204 A

Space for rough work

5. Equilibrium constant for the dissociation of $Ag(NH_2)_2^+$ into Ag^+ and NH_3 is 6×10^{-8}. Calculate $E°$ for the following half reaction, $Ag(NH_3)_2^+ + e \rightarrow Ag(s) + 2NH_3$

$\left(E^\circ_{Ag^+/Ag} = 0.799V, \log 6 = 0.7781\right)$

(A) 0.37
(B) 0.799
(C) 0.7781
(D) 0.6

6. A student is analysing a salt in the laboratory. When he added dilute HCl to the salt solution, a gas is evolved, when the gas is exposed to a filter paper moistened with potassium iodate and starch solution, he observed blue colouration of the filter paper. Then he added $BaCl_2$ solution to the salt solution and observed white precipitate immediately. He found that the precipitate is soluble in dilute mineral acids. The anion of the salt is

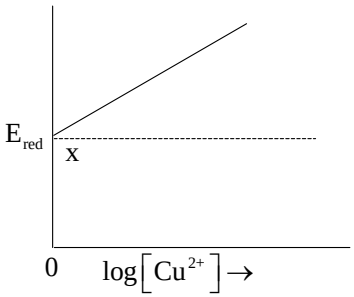
(A) $S_2O_3^{2-}$
(B) SO_3^{2-}
(C) CO_3^{2-}
(D) HCO_3^-

7. Which of the following complexes is/are chiral?

(A) $cis-[Pt(NH_3)_2Cl_2]$
(B) $[Zn(H_2O)_2Cl_2](\text{tetrahedral})$
(C) $\left[Pt(C_2O_4)_3\right]^{4-}$
(D) trans - $\left[Pt(NH_3)_4BrCl\right]^2$

8. For the reaction $Cu^{2+} + 2e \rightarrow Cu$

$\log\left[Cu^{2+}\right]$ vs. E_{red} graph can be plotted as following (At NTP)



Where OX = 0.34 V then electrode potential of the half cell of $Cu/Cu^{2+}(0.1M)$ will be:

(A) 0.3695 v
(B) − 0.3105 V
(C) 0.3991 V
(D) −0.2809 V

Space for rough work

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

9. An ideal gas expanding isothermally against the external pressure equal to zero, shows
   (A) $\Delta E = 0$ (B) $W = 0$
   (C) $q = 0$ (D) $\Delta H = +ve$

10. Electrolysis of molten $MgCl_2$ is the final production step in the isolation of magnesium from sea water by Dow's process. Assuming that 35.6 g of Mg metal forms. Which of the following statement (s) is/are correct?
   (A) 2.96 moles of electrons are required
   (B) $3.83 \times 10^5 C$ charge is required
   (C) 31.8 A of current is required for 2.5 hours
   (D) 41.5 A of current is required for approximately 115 mins

11. A 10 mL mixture containing 0.2 g NaOH and $0.25 g\ Na_2CO_3$ was treated with $x N\ HCl$ required 15 mL of it with phenolphthalein indicator. After end point a $0.5\ g\ NaHCO_3$ was added and it was again titrated with $y\ N\ HCl$ required 25.5 mL of HCl with methyl orange. Which of the following is/are correct?
   (A) x is almost 1.5 times of y
   (B) If in above titration x N HCl is replaced with y N HCl then for first end point it will require 23mL of HCl.
   (C) After first end point instead of 0.5 g if 2.5 g $NaHCO_3$ is added and fitrated with $x\ N\ HCl$ it will require 65.5 mL.
   (D) Only (A) and (B) are correct.

12. In which of the following pair the later species is not having higher vapour pressure at a given temperature than former one?
   (A) C_2H_6 and C_4H_{10} (B) CH_3CH_2OH and CH_3CH_2F
   (C) Xe and I_2 (D) NH_3 and PH_3

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 13 & 14

A hydrated metallic salt A, light green in colour, on careful heating gives a white anhydrous residue B. B is soluble in water and its aqueous solution reacts with NO to give a dark brown residue D and on heating the mixture it gives two gases E and F.

13. The gaseous mixture (E + F) when passed through acidified permanganate solution, discharges the pink colour and when passed through acidified $BaCl_2$ solution gave a white precipitate Identify the gasses E and F.

(A) CO_2 and CO (B) NO_2 and N_2O

(C) NO and NO_2 (D) (D) SO_2 and SO_3

14. A hydrated metallic salt A and its anhydrous form B respectively are:

(A) $CuSO_4.5H_2O$ and $CuSO_4$ (B) $FeSO_4.7H_2O$ and $FeSO_4$

(C) $CaSO_4.2H_2O$ and $CaSO_4$ (D) $CaSO_4.1/2H_2O$ and $CaSO_4$

Space for rough work

Paragraph for Questions 15 & 16

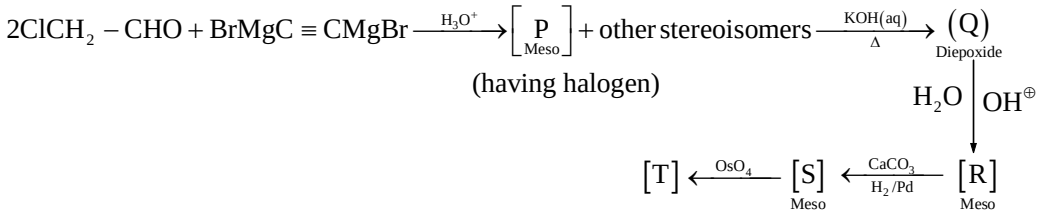
Strength of the sample of H_2O_2 is generally expressed in terms of the volume strength. It means the volume of oxygen liberated at N. T. P. by heating one volume of hydrogen peroxide. The concentration of H_2O_2 in a solution can also be expressed as percentage of H_2O_2 in solution. Normality of this solution can be calculated if the equivalent mass of H_2O_2 is known.

15. 25 ml of H_2O_2 solution was added to the excess of acidified KI solution. The iodine so liberated required 40 ml of 0.1 N sodium thiosulphate solutions. Calculate the normality of H_2O_2 solution.
(A) 0.08 (B) 0.16
(C) 0.02 (D) 0.20

16. The volume strength of the H_2O_2 solution is:
(A) 0.556 (B) 0.448
(C) 0.896 (D) 0.632

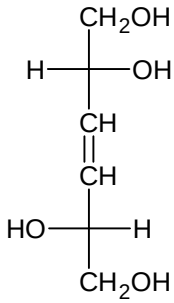
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Paragraph for Questions 17 & 18

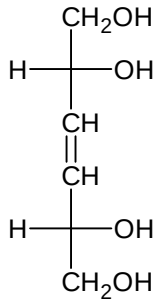


17. Compound S is?

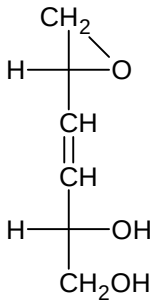
(A)



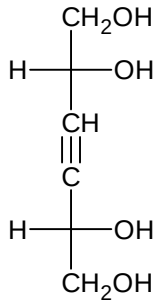
(B)



(C)

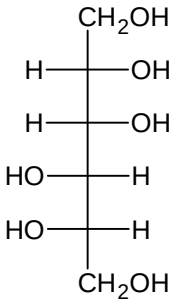


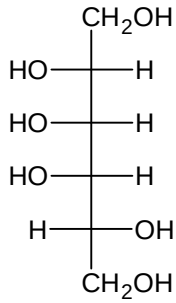
(D)

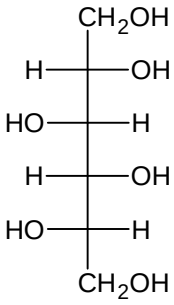


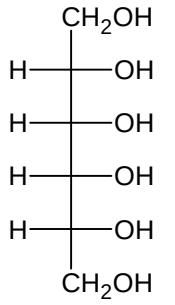
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18. Compound T is?

(A) 

(B) 

(C) 

(D) 

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

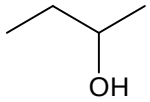
1. Find the number of native ores out of given ores?
Pyrolusite,Chromite,Siderite,Cassiterite,Calamine,Argentite,Limestone,Chalcopyrite.

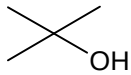
2. Out of given chemical, number of chemicals which is/are related to Solvay process:
$NH_3, KCl, CO_2, SO_2, Ca(OH)_2, NaCl, H_2SO_4, NaNO_3$:

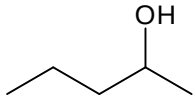
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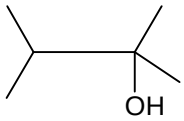
3. Find the number of properties which is/are correctly related to graphite?
(i) thermodynamically most stable allotropes of carbon
(ii) highest thermal conductivity among all allotropes of carbon
(iii) lowest electrical resistance among all allotropes of carbon
(iv) C-C covalent bond length is higher as compared to diamond
(v) hexagonal rings in structure
(vi) football like molecule

4. Consider the following 3d series ions:
$Sc^{3+}, Ti^{3+}, V^{3+}, Cr^{3+}, Mn^{3+}, Fe^{3+}, Co^{3+}, Ni^{3+}$
Find total number of ions for which all octahedral complexes are paramagnetic, irrespective of ligand

5. (a)  $\xrightarrow[\Delta]{H^+}(x)$

(b)  $\xrightarrow[\Delta]{H^+}(y)$

(c)  $\xrightarrow[\Delta]{H^+}(z)$

(d)  $\xrightarrow[\Delta]{H^+}(p)$

Total number of products obtained in above reactions including minor products is (including stereoisomer including stereoisomer)

Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **08 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which only **ONE** is correct

1. The ordered pair (a, *b*) such that $x^2 - x - 1$ divides the polynomial $ax^9 + bx^8 + 1$ is
   (A) (–21, 34) (B) (21, –34)
   (C) (47, –58) (D) (58, –47)

2. The number of possible integral pairs (*a*, *b*) for which $2ab - 5a + b = 55$ is
   (A) 8 (B) 16
   (C) 12 (D) 6

3. If $f_n = \sum_{k=0}^{n} (1 + 2^k)\binom{n}{k}$, then
   (A) $f_{n+2} = 3f_{n+1} - 2f_n$ (B) $f_{n+2} = 3f_n + 2f_n$
   (C) $f_{n+2} = 5f_{n+1} - 6f_n$ (D) $f_{n+2} = 6f_{n+1} - 5f_n$

4. A set contains 3*n* members. Let P_n be the probability that *S* is portioned into 3 disjoint subset with *n* members in each subset such that the three largest numbers in *S* are in different subsets. Then $\lim_{n\to\infty} P_n$ is
   (A) $\frac{2}{7}$ (B) $\frac{1}{7}$
   (C) $\frac{1}{9}$ (D) $\frac{2}{9}$

Space for rough work

5. If a, b, c are in H.P. and p is the length of the perpendicular drawn from the origin to any member of family of lines $(a + c - 2ac)\,bx - 2acy + 7abc = 0$ then $p \in$

(A) $\left[\frac{1}{4}, \frac{9}{\sqrt{2}}\right]$ (B) $\left(0, \frac{7}{\sqrt{2}}\right]$

(C) (7, 9) (D) $\left[\frac{7}{\sqrt{2}}, \frac{9}{\sqrt{2}}\right]$

6. The sides of a triangle are the roots of the cubic equation $x^3 - 13x^2 + 54x - 72 = 0$ then the area of the triangle in square units is

(A) $\frac{1}{2}\sqrt{455}$ (B) $\frac{1}{4}\sqrt{455}$

(C) $\frac{1}{2}\sqrt{394}$ (D) $\frac{1}{4}\sqrt{394}$

7. f(x) is a polynomial function and $(f(\alpha)^2 + (f'(\alpha))^2 = 0$ then find $\lim_{x\to\alpha} \frac{f(x)}{f'(x)}\left[\frac{f'(x)}{f(x)}\right]$ where [.] denotes greatest integer function.

(A) 0 (B) 1

(C) –1 (D) none of these

8. f(x) and g(x) are quadratic polynomials and $|f(x)| \geq |g(x)|, \forall x \in R$. Also f(x)=0 have real roots. Then number of distinct roots of equation $h(x)h''(x) + (h'(x))^2 = 0$ are (where h(x)=f(x)g(x)).

(A) 0 (B) 2

(C) 3 (D) 4

Space for rough work

Multiple Correct Choice Type

This section contains **04 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct

9. If the equation $px^2 + y^2 + qz^2 + 2yz + zx + 3xy = 0$ represents a pair of perpendicular planes, then $p - q =$

(A) 5 (B) –5

(C) $\frac{5}{2}$ (D) $-\frac{5}{2}$

10. If $\frac{dy}{dx} + y\frac{dx}{dy} = x$ and $y(-2) = 1,$ then

(A) $2x + y + 3 = 0$ (B) $x^3 + 8y = 0$

(C) $x + y + 1 = 0$ (D) $x^2 - 4y = 0$

11. If $(1, 2, p)$, $(2, 2p, -6)$ and $(\alpha^2 - 2\alpha, 1, 1)$ are the ordered triplets of form (*x, y,* z) which satisfies all the equations.

$\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1,$ $\frac{x}{b} + \frac{y}{c} + \frac{z}{a} = 1$ and $\frac{x}{c} + \frac{y}{a} + \frac{z}{b} = 1$ then α can be

(A) 2 (B) – 2

(C) – 4 (D) 4

12. If $f(x)$ is a continuous and differentiable function such that $f(t_k) = 1$ for $k = 1, 2, ...n,$ where $t_k = \sum_{r=1}^{k} \frac{1}{r(r+1)(r+2)}$ then

(A) $f\left(\frac{1}{4}\right) = 1$ (B) $f\left(\frac{1}{2}\right) = 1$

(C) $f'\left(\frac{1}{2}\right) = 0$ (D) $f'\left(\frac{1}{4}\right) = 0$

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct

Paragraph for Questions 13 & 14

Consider the function $h(n) = \int_{o}^{a} f(x)\, g(nx)\, dx$

13. Let $f(x) = 2,\ g(x) = |\sin \pi x|,$ and $n = \frac{1}{N},$ where *N* is natural number. If $h(n)/a$ is independent of *n* for $N = 1, 2, 3, 4$ then minimum value of *a* is
(A) 2 (B) 3
(C) 4 (D) 12

14. If $f(x) = \sin^2 x,\ g(x) = |\cos x|$ then $\lim_{a\to\infty} \frac{h(1)}{a}$ is equal to
(A) $\frac{2}{3\pi}$ (B) $\frac{1}{3\pi}$
(C) $\frac{1}{2\pi}$ (D) $\frac{1}{\pi}$

Paragraph for Questions 15 & 16

In a $\Delta ABC,\ AB = 3,\ BC = 4,\ CA = 5.$ *P* is any point inside the $\Delta ABC,$ such that $\angle PAB = \angle PBC = \angle PCA = \alpha.$

15. $\cot\alpha =$
(A $\frac{1}{5}$ (B) $\frac{25}{4}$
(C) $\frac{25}{12}$ (D) 4

Space for rough work

16. Ratio of Areas of ΔPBC and ΔPAB is

(A) $\frac{16}{25}$ (B) $\frac{25}{16}$

(C) 2 (D) $\frac{4}{15}$

Paragraph for Questions 17 & 18

Using three basic colours (Red, Blue, Green) different colour are made by mixing these two or more colours in equal proportion. The *n* vertical stripes are painted by all colours available, such that no two consecutive stripes have same colour.

17. The number of ways of painting *n* stripes such that only alternate stripes are painted by a basic colour (*n* in odd) is

(A) $3^{\frac{n+1}{2}}.4^{\frac{n-1}{2}}$ (B) $7.12^{\frac{n+1}{2}}$

(C) $3^{\frac{n-1}{2}}.4^{\frac{n+1}{2}}$ (D) $7.12^{\frac{n-1}{2}}$

18. If *n* stripes are painted such that no two consecutive stripes have any basic colour common, then number ways of pointing *n* stripes is

(A) $\frac{1}{2}\left(\left(1+\sqrt{2}\right)^{n-1}+\left(1-\sqrt{2}\right)^{n-1}\right)$ (B) $\frac{1}{\sqrt{2}}\left(\left(1+\sqrt{2}\right)^{n-1}-\left(1-\sqrt{2}\right)^{n-1}\right)$

(C) $\frac{1}{2}\left(\left(1+\sqrt{2}\right)^{n}+\left(1-\sqrt{2}\right)^{n}\right)$ (D) $\frac{3}{\sqrt{2}}\left(\left(1+\sqrt{2}\right)^{n}-\left(1-\sqrt{2}\right)^{n}\right)$

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. Let L_1 be the projection and L_2 be the image of the z-axis in the plane $3x-4y+z+1=0$. The distance of the point (1, 2, 3) from the plane containing the lines L_1 and L_2 is ……

2. The graphs of $x^2+y^2+6x-24y+72=0$ and $x^2-y^2+6x+16y-46=0$ intersect at four points. The sum of distances from these four points to the point $(-3,\ 2)$ is 10 *k*. Then value of *k* is equal to

3. Let $A=[a_{ij}]_{n\times n}$ be a matrix such that $a_{ij}=i.2^{-j}$ then find the value of $\lim\limits_{n\to\infty}\left(\text{trace } A^n\right)^{1/n}$ (where $n\in N$).

4. $f(x+1)=(-1)^{x+1}x-2f(x)$ for $x\in N$ and $f(1)=f(1986)$.
Compute the sum of digit of number $\left(f(1)+f(2)+\ldots f(1985)\right)$

5. Two rays begin at point *A* and form an angle of 43º with one another. Lines l_1, l_2, l_3 (no two being parallel) each form an isosceles triangle with the original rays. If θ^o is the largest angle of triangle formed by l_1, l_2 and l_3 then find value of $8\cos^2\left(\theta^o-9^o\right)$.

Space for rough work